

Rohithkumar Saravanan

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Skills

Languages: Python, Matlab, C/C++, JavaScript, SQL, HTML/CSS

Technical: Machine Learning, Deep Learning, Computer Vision, Image Processing, Git/GitHub, Matlab, Front-End Development

Non-Technical: Communication, Project Management, Leadership, Team Work, Time Management, Adaptability

Education

University of Limerick

Master's in Artificial Intelligence and Machine Learning (Ongoing)

SRM Institute of Science and Technology

B.Tech in Electronics and Computer Engineering

CGPA: 8.34

Experience

Python Intern

Aug 2023 – Nov 2023

Shiash Info Solutions Private Limited

- Completed a hands-on internship focused on Python programming and real-world problem solving.
- Worked on the project “**Estimation of Carbon for Industry and Household to Prevent Environment and Industry Progression**”.
- Developed Python-based models to estimate carbon emissions from industrial and household sources.
- Applied data analysis techniques to support environmental sustainability and industrial efficiency.

AI-ML Virtual Intern

Apr 2024 – Jun 2024

Google for Developers (India Edu Program)

- Gained foundational knowledge of Artificial Intelligence and Machine Learning through a structured 10-week program.
- Worked on ML model training and optimization techniques to improve accuracy and performance.
- Explored best practices within the Google Developer ecosystem to design scalable AI solutions.

Cyber Security Intern (Industrial Training)

Mar 2023 – Apr 2023

Industrial Training Program

- Completed industry-mentored training focused on Cyber Security fundamentals.
- Gained exposure to threat analysis, system vulnerabilities, and security best practices.

Projects

1. **Advanced Surface EMG Analysis for ALS Neuromuscular Dysfunction**

March 2025

- Developed Arduino-based sEMG acquisition hardware.
- Built preprocessing pipeline and deep learning models (CNN, LSTM, CNN-LSTM).
- Designed web UI for real-time prediction and visualization.

2. **Enhancing Chest CT Scan Infection Type Classification using BEMD**

April 2024

- Proposed new preprocessing method for enhanced CT scan imaging.
- Built DL model for multi-class infection classification.

3. **Pneumonia Detection using PDNN**

July 2024

- Developed a dropout-regularized neural architecture for pneumonia detection in CT scans.

4. **Biorthogonal Wavelet Image Enhancement for Pneumonia Diagnosis**

August 2024

- Built wavelet-based enhancement pipeline for CT images.
- Achieved 99.20% accuracy using DenseNet.

5. **Deep Transfer Learning for Lung Cancer Classification**

July 2024

- Applied GoogLeNet with LIDC dataset for cancer detection.
- 6. **Triplet-Attention Residual Network for Pneumonia Detection** July 2024
 - Developed LuNet achieving 95.34% accuracy on CT datasets.
- 7. **Home Automation using NodeMCU** March 2023
 - Smart socket controlled via Alexa, Google Assistant, and manual switches.
- 8. **Multifunction Robot using Arduino** November 2022
 - Built voice-controlled and Bluetooth-controlled robot with obstacle avoidance.

Awards

- Best Paper Award – All India Two Days Seminar** July 2024
 - Organized by The Institution of Engineers (India).
- Best Project Award – Multidisciplinary Domain** April 2025
 - SRM Institute of Science and Technology.
- 3rd Place – COMSPRO 2025 Project Contest** April 2025
 - SRM Institute of Science and Technology.

Publications

1. **Biorthogonal Wavelet-Based Image Enhancement for Accurate Pneumonia Diagnosis from Lung CT Scans Using Deep Learning.** Springer Link
2. **Advanced Surface EMG Analysis for ALS Neuromuscular Dysfunction Using Deep Learning Models** (Submitted)

Workshops and Seminars

- Signal Processing and Artificial Intelligence** April 2023
 - Organized by The Institution of Engineers (India)
- Kuka Robotics Workshop** September 2023
 - Organized by Chennai Institute of Technology
- Industrial Automation Workshop** October 2023
 - Organized by Chennai Institute of Technology
- All India Two Days Seminar** July 2024
 - The Institution of Engineers (India)

Certification Courses

1. Supervised Machine Learning (Coursera)
2. Advanced Computer Vision with TensorFlow (Coursera)
3. Advanced Diploma in Python Programming (CSC Institute)
4. Introduction to Front-End Development (Coursera)
5. Programming with JavaScript (Coursera)
6. HTML and CSS in Depth (Coursera)
7. Version Control with Git (Coursera)
8. Matlab Onramp (MathWorks)
9. Arduino Programming (KitFlix)

Position of Responsibility

- Committee Head – Directorate of Student Affairs (DSA)** June 2021 – May 2025